Exercises 09/08

1. Define $\mathbb{R}^n = \mathbb{R} \oplus \mathbb{R}^{n-1}$ for $n > 2$. Show $\mathbb{R}^n$ is a vector space under coordinate-wise operations.
2. Consider the set $\mathbf{W}$ of vectors in $\mathbb{R}^2$ whose components sum to 1, i.e., $\mathbf{w} \in \mathbf{W}$, where $\mathbf{w} = (w_1, w_2)$ such that $w_1 + w_2 = 1$. Is $\mathbf{W}$ a vector subspace?
3. Is $\{(1,0,0),(0,1,0),(1,1,0)\}$ linearly independent?
4. Show that any set containing $\mathbf{0}$ is linearly dependent.
5. Let $B = \langle \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \rangle$ and $\mathbf{v} = (1,1,0)$

   a. Represent $\mathbf{v}$ in terms of B.

   b. Verify the Exchange Theorem.
6. Is $f(x) = x^2$, a linear map?
7. Show that if $f : \mathbf{V} \to \mathbf{W}$ is a linear map then $f(0) = 0$.
8. Let $f : \mathbf{V} \to \mathbf{W}$ be a linear map. Show that $\ker(f)$ is a subspace of $\mathbf{V}$ and $f(V)$ is a subspace of $\mathbf{W}$
9. Suppose $\mathbf{V} = \mathbb{R} \oplus \mathbb{R}$ and $\mathbf{W} = \mathbb{R}$. Define $f : \mathbf{V} \to \mathbf{W}$ by $(v_1, v_2) \to v_1 + 2v_2$. Find a basis for the kernel of f.